THE WATNEY LOG: SSAC 2026 HACKATHON MASTER REFERENCE

One document to rule them all. One document to find them. One document to bring all the insights and in the darkness bind them into a winning presentation.

"I'm going to have to science the shit out of this."

What this is: Everything you need to know for the March 5th SSAC Hackathon, consolidated from seven industry reports, three Wasserman studies, two uploaded domain guides, one Tucker sync, and approximately 47 hours of Claude-assisted obsession. If it's not in this document, you don't need it. If it IS in this document and you didn't read it, that's on you, astronaut.

How to use it: Three passes.

- **Pass 1 (skim):** Read bolded lines and tables only. Takes 15 minutes. Gets you 70% there.
- **Pass 2 (absorb):** Read the full sections. Takes 45 minutes. Gets you to 95%.
- **Pass 3 (apply):** Quiz yourself with the flashcards at the end. If you can answer 80%, you "know hockey."

PART 1: THE MISSION BRIEFING

What Is This Hackathon

The 13th annual SSAC Hackathon, presented by Wasserman Foundation. Thursday, March 5th, 2026, Boston. You arrive at 9 AM. You receive a dataset and problem statement you've never seen. You have approximately 7 hours to analyze it, build something, and present it to a panel of industry judges. Top teams advance to present on the main conference stage March 6–7.

Teams of two maximum. You're partnered with **Tucker Paron** — ML Engineer at Neurable, former Data Scientist at Spurs Sports & Entertainment, and an **SSAC 2024 Hackathon Finalist.** He's been in the room. He knows the judges. He knows the pacing. This is not a small advantage.

The Theme

"Unlock the value of fandom" in women's sports.

The official language: *"This challenge looks to reveal the true commercial potential of an audience through the analysis of first party data. The aim is to go beyond traditional 'revenue per fan' models for merchandise, membership, tickets and hospitality — by uncovering new partnership models."*

Three layers to decode:

Layer	What it says	What it means
Surface	"Unlock fandom value"	Women's sports fans are undermonetized
Middle	"Beyond revenue per fan"	Don't just count ticket sales. Find NEW value.
Deep	"New partnership models"	Psychographic segmentation → sponsor matching → activation strategy

The judges are likely Wasserman executives. They've spent seven years researching this exact topic through their subsidiary, The Collective. **They already know the answer they want to hear.** Your job is to prove it with their data, add a novel angle, and present it so cleanly they want to hire you.

The Dataset You'll Probably Get

Based on the 2025 hackathon (also Wasserman-sponsored, which provided "tens of thousands of rows of anonymized first-party fan data" from a global football club) and Wasserman's published research methodology, expect:
Format: CSV, 10,000–50,000 rows, anonymized first-party fan data. Likely from a WNBA, NWSL, or multi-property women's sports source.
Probable columns:

Category	Likely Variables
Demographics	Age/generation, gender, geography, income bracket, household composition, motherhood status
Behavioral	Game attendance frequency/recency, merch purchases, ticket spend, digital/app engagement, content consumption
Commercial	Sponsor awareness, purchase intent, brand affinity scores, partner interaction data
Psychographic	Fandom intensity/avidity, values alignment, motivations for following, household decision-maker status

What's probably NOT in there: Raw ticket transaction pricing, play-by-play data, athlete social media raw feeds. This is survey + behavioral, not transactional.
Key implication: Prepare for mixed data types — categorical demographics alongside ordinal Likert scales alongside continuous spending variables. Your clustering approach needs to handle all three.

PART 2: THE WOMEN'S SPORTS UNIVERSE IN 20 MINUTES

The Rocket Model

ThiLnk of women's sports monetization as a rocket:

Component	What it represents	Key reports
FAN FUEL	Who fans are, how they behave, what they buy	The Collective "Her Fandom", New Faces (iso/duo-fans)
ENGINE	Athletes + teams generating influence and engagement	New Economy Part I
THRUST	Media rights, sponsorships, merch, tickets	McKinsey, Deloitte, SponsorUnited
ORBIT	Valuations + investment ecosystem	New Economy Parts II & III

Everything connects. Fan behavior drives sponsor ROI. Sponsor ROI drives media rights pricing. Media rights drive valuations. Valuations attract capital. Capital improves infrastructure. Better infrastructure attracts more fans. It's a flywheel, and it's spinning faster than anyone predicted.

The Big Picture: Where the Money Is

Global women's sports revenue nearly doubled from **$981M in 2023 to $1.88B in 2024**, with projections hitting **$2.35B in 2025.** That's approximately **4.5x the revenue growth rate of men's sports** over the same window. This isn't a blip. This is a market correction.

Revenue breakdown (2025 projected):

Revenue Stream	Amount	Share
Commercial (sponsorship/merch)	$1.26B	54%
Broadcast/media rights	$590M	25%
Matchday (gate/concessions)	$500M	21%
Total	**$2.35B**	**100%**

The critical insight: Sponsorship is carrying the load, not media rights. In men's sports, broadcast is the dominant revenue driver. In women's sports, commercial partnerships are 54% of revenue. This means the brands are ALREADY there — the question is whether they're reaching the right fans with the right activations. That's literally what the hackathon is asking you to solve.

North America dominance: 56% of global revenue in 2024, projected to hit 59% ($1.39B) in 2025. This is your operating market.

The Monetization Gap (Why This Hackathon Exists)

McKinsey's headline: women's sports is a **$2.5B media rights opportunity by 2030** but currently represents **under 2% of the ~$75B U.S. sports market.**

The gap isn't shrinking — it's actually getting worse in one specific dimension:

Metric	Men's Sports	Women's Sports	Gap
Media revenue per viewing hour	$1.25	$0.25	5x
Media rev/hour CAGR (2017–2023)	+7%	**-3%**	Diverging
NCAA March Madness 30-sec ad spot	$2M (14.8M viewers)	$500K (18.9M viewers)	4x for FEWER viewers
Share of U.S. sports market	98%+	<2%	49x

Read that NCAA line again. The women's final had **more viewers** and the ad spot cost **4x less.** That's not a valuation model. That's a market failure. And market failures are where money gets made.

WNBA media deal repricing: New 11-year deal starting 2026, worth ~$2.2B total ($200M/year). That's **3x the prior deal** and growing **2.6x faster** than the NBA's recent renewal. The NWSL broadcast deal went up **40x.** The market is correcting — fast, but not fast enough.

The Fan: Who Actually Watches Women's Sports

This is the hackathon's core territory. "Revenue per fan" is a blunt instrument. The real value is in understanding WHO these fans are and WHY they engage.

The Decision-Maker Problem

From Wasserman's "Her Fandom, Her Buying Power" study (7,100+ women, 10 countries):

- **84% of women sports fans are key decision-makers in sports-related purchases** (48% primary + 36% shared)
- **66% say sports organizations don't understand or appeal to them**

- **39% say brands don't recognize them as sports fans**
- Women will control **75% of discretionary spending by 2030** and lead **85% of household purchasing decisions**
- Among mothers: **91% are actively involved in kids' sports** (tied to $77B youth sports industry)

The one-sentence pitch: **"Women aren't just 'audience' — they're the household sports CFO, and the industry keeps marketing like the CFO doesn't exist."**

The Time-Starved Fan

U.S. women have **13% less free time** than men. This gap widens for mothers. Fandom that requires digging for schedules, juggling subscriptions, and hunting highlights is taxing the scarcest resource on earth: time. Women fans are described as **"story-driven consumers"** who engage more with behind-the-scenes content, docuseries, and athlete narratives than pure highlights or scores.

What this means for the hackathon: If there's content consumption data, the analytical lens is "what kind of engagement survives time scarcity?" Short-form, mobile, values-aligned, on-demand. Not "tune in at 7 PM or miss it."

The Three Fan Archetypes (Wasserman's New Faces Report)

This is your potential "wow factor." Most teams will segment by demographics. You're going to segment by **social consumption pattern.**

Segment	Prevalence (Women's Sports)	Prevalence (Men's Sports)	Key Signal
IsoFan	18–21%	5.6%	Consumes alone. Low social media activity. High spend potential.
DuoFan	25–28%	N/A	Shares fandom with one person (spouse/friend).
Social Fan	54–59%	80%+	Communal engagement. Supporter groups. Discord.

Why iso-fans are a goldmine:

- They're 3x more prevalent in women's sports than men's
- **46% wish they had more people to share fandom with**
- **62% would join a fan group if made more accessible**
- They're invisible to traditional sports marketing (no arena attendance, no tweets, no group tickets)
- But they may be spending heavily on streaming, merch, and digital content — alone, at home, at 11 PM
- This isn't a sad story. It's a **product and marketing opportunity.** The community infrastructure simply doesn't exist yet for women's sports the way it does for men's.

The 50% identity stat: Half of women's sports fans identify as fans of **women's sports generally**, not a specific team (37%) or league (13%). This means athletes and storylines are the acquisition hooks, not team loyalty. "Women's sports pass" bundles make more sense than team-specific packages.

Movement-Based Fandom

This isn't just entertainment consumption. It's identity alignment.

- Women's sports fans are **2.1x more likely** to consider a brand because of its sponsor association than men's sports fans
- **50% of women's sports fans** "strongly agree" sponsors should make the world a better place (vs. 20% of men's sports fans)
- **32% favor environmentally conscious brands** (vs. 29% of men)
- **89% are more inspired to take social action** when brands partner with women athletes

The framing: Purpose-driven loyalty. These fans aren't just buying merch — they're voting with their wallets for the world they want. Sponsors who align with these values unlock loyalty that traditional sports marketing can't touch.

The Generational Split

Generation	Avidity	Commercial Receptivity	Key Behavior
Gen Z	Rising fast	Highest — 46% more likely to buy from sponsors	Creator-style content, athlete-led commerce, social proof
Millennial	Highest (3 in 4 are avid)	High — most likely to attend live events, most receptive to sponsorship	Peak engagement, peak spend
Gen X	Moderate	Moderate	Traditional consumption patterns
Boomer	Declining (59.6% avid vs. 78.8% Gen Z)	Lower	Less digitally active

77% of Gen Z trust that women athletes genuinely believe in the products they endorse. That's an authenticity premium that men's sports endorsements don't get.

The Athlete Value Arbitrage

From Wasserman/RBC "New Economy of Sports" Part I:

- Men earn **~21x more in playing salary** than women athletes (average)
- Women depend on sponsorship/endorsement income **~2x more** than men (because base pay is lower)
- **90% of partnership dollars** go to men's athletes
- But women athletes drive **2x more fan engagement** per dollar spent
- **20% of consumers** follow individual women athletes on social media (vs. 18% who follow teams, 10% who follow leagues)

The mispricing thesis: The market pays for reach (audience size). Women athletes deliver on **outcomes** (trust, engagement, purchase intent, social action). The compensation model is broken because it measures the wrong thing. If your hackathon analysis can quantify this at the fan level, you're speaking Wasserman's exact language.

The Sponsorship Landscape

Sponsorship is already working. The question is how to make it work harder.

Metric	Women's Sports	Men's Sports
Brand recall effectiveness	2x	Baseline
Purchase intent lift	+10%	Baseline
Fans likely to try sponsor products	75% (NWSL)	64% (MLS)
Fans supporting league sponsors	58% (WNBA)	45% (NBA)
Consumer favorability toward sponsors	30%	20%

WNBA team sponsorship revenue: $76M in 2024, sponsor value up **312% YoY. NWSL total sponsorship revenue:** $75M, with Angel City FC outperforming 15 MLS and 25 MLB teams per-game. **WNBA Changemaker partners:** 286% ROI on average. **Ally Financial's 50/50 pledge** (equalizing men's/women's media spend): 27% increase in brand preference, 40% rise in brand consideration within 10 months. **Non-endemic brands entering:** Beauty (Sephora × Valkyries, Fenty × Liberty), healthcare, finance. 82% of Gen Z sports fans identify as "beauty enthusiasts" — a connection unexplored until recently.

PART 3: THE CASE STUDIES (YOUR OUTLIER FRAMING)

You and Tucker agreed the "outlier" angle is your narrative device. Here are the outliers you need to know cold.

Angel City FC: The Revenue Standard

Angel City inverted the traditional sports business model. They didn't wait for media rights to fund the team — they built a community-first, values-first, data-first operation.

Metric	Number	Context
Valuation	$250–$280M	Highest of any women's soccer team globally
Inaugural season sponsorship	$20M	Outperformed 15 MLS and 25 MLB teams per-game
Revenue per social follower	$29	vs. $13 for LA Dodgers, $8 for NY Yankees
First-season merch	$6M	38% of entire NWSL league merch revenue
Community reinvestment	$7M (10% model)	Fans attend out of "gratitude and community connection"
Fan database growth	300%+ in 2 years	CRM-driven, unified all touchpoints via HubSpot

The "48-hour window": Angel City discovered fans are exponentially more likely to purchase within 48 hours of a match. By automating personalized push notifications during this period, they turn game-day emotion into next-day revenue.

Golden State Valkyries: The Expansion Blueprint

The proof that with proper capital and strategy, you can build a commercially dominant women's sports franchise from scratch.

Metric	Number	Context
Expansion fee	$50M	Paid by Joe Lacob and Peter Guber
Current valuation	~$500M	10x increase in under 2 years
Home game attendance	18,064 avg (all 22 sold out)	Higher than 15 NBA teams in 2024–25
Pre-season ticket sales	10,000+ season tickets	Before playing a single game

Crossover with Warriors fans	<5% overlap	Acquired a genuinely NEW audience

The "Bright Believers": Valkyries' core fan segment — affluent, young (20–35), purchase based on lifestyle curation, not traditional sports allegiance. Sephora is a founding partner with naming rights to the practice facility. This is what "new partnership models" looks like.

Nebraska Volleyball: Grassroots Commercial Power

The proof that "non-revenue" women's sports can drive mass-market economics.

Metric	Number	Context
Consecutive sellout streak	306 matches	At Devaney Center
2023 stadium match attendance	92,003	World record for women's sports attendance
Stadium match single-night revenue	$1.4M	Filled nearly every hotel room in Lincoln
FY2024 total revenue	$7.3M	Program profit: +$1.3M
FY2025 revenue	$5.6M	Deficit: -$1.38M (costs professionalized faster)

The cautionary note: Nebraska's FY2025 deficit ($1.38M) shows the other side — as women's sports scale, so do costs (chartered flights $1M, coaching bonuses $410K). Commercial innovation must keep pace with professionalization. This is a mature nuance that will impress judges.

UConn Women's Basketball

The dynasty model. UConn recently got legislation passed allowing up to $20.5M annually in revenue-sharing with student-athletes, professionalizing the collegiate pipeline. Their brand value, built over decades of dominance, is now being converted into direct commercial infrastructure.

PART 4: THE ARGUMENTS AGAINST (KNOW THEM SO YOU CAN BEAT THEM)

Smart teams don't just argue their case — they preemptively defuse the counterarguments. If a judge plays devil's advocate, you want to be ready.

"Women's leagues lose money." True historically. FIFA's 2024 report found 67% of "tier one" women's football clubs are loss-making, with average player salary of $10,900. But this describes a pre-infrastructure, pre-media-deal era. The WNBA's new $200M/year deal and NWSL's 40x broadcast increase are structural corrections, not flukes. WUSA (2003) and WPS (2012) collapsed in a pre-digital, under-capitalized environment that bears almost no resemblance to today's landscape.

"Valuations are inflated." WNBA teams trade at 14.4x revenue vs. 11.7x for NBA. Yes, current prices are a bet on future growth. But expansion fees have increased 5x in ~20 months, and the Valkyries validated the bet immediately ($50M fee → $500M valuation in under 2 years).

"Media coverage is still 5%." ESPN's SportsCenter still devotes ~5% of airtime to women's sports despite 78% of consumers wanting equal coverage. This is "institutional under-marketing" — and it's a gap to exploit, not a ceiling to accept.

"The political angle." Tucker flagged this: the "everybody watches women's sports" movement is politicized. Keep this as context you understand but don't lead with. If the data reveals values-driven fandom, frame it as "values-driven loyalty" — same insight, commercially actionable language, zero judge alienation risk.
Your response formula: "The skeptics point to [X]. Our data shows [Y], and here's the $[Z] they're leaving on the table." Always end on the opportunity, never on the objection.

PART 5: THE HACKATHON STRATEGY

What Wins (Lessons From Past Champions)

Ben Jenkins — only back-to-back winner (Open Division 2024 + 2025). Won both times with a partner, including the 2025 Wasserman-sponsored hackathon analyzing fan data from a global football club.
Daniel Lee — 2022 winner, wrote the best public post-mortem. His five principles:

1. **Define the problem before touching data.** First 30 minutes = whiteboarding, not coding.
2. **Sound like you know the domain.** "It sounded like we knew hockey" was the compliment that told him he won.
3. **Build something interactive.** Judges remember experiences, not slides.
4. **Polish the presentation ruthlessly.** Cut everything that doesn't serve the story.
5. **Scope simply but novel.** Answer one question nobody else has answered.

Judging criteria (from SSAC): Creativity, functionality, originality, innovativeness. Note: "functionality" means it works and is useful, not that it's technically complex.
What separates winners from everyone else: Winners lead with the conclusion, demonstrate deep domain knowledge, and make specific business recommendations with dollar figures. Everyone else saves their best insight for slide 8, presents methodology instead of insights, and runs out of time on the presentation because they were still coding at 3 PM.

The Khalid + Tucker Strategy (From Your Sync)

Three-layer approach:

1. **The premise:** Wasserman believes women's sports fandom has untapped commercial value. Prove it with data. Don't just agree — quantify.
2. **The method:** Psychographic/behavioral segmentation over demographics. Identity, motivations, cross-pollinated behaviors tied to commercial activation.
3. **The presentation:** One sharp, novel story beats broad coverage.

Outlier framing: Anchor around breakout properties (Nebraska, Angel City, Valkyries, UConn). What makes their fandom work commercially? Can you find those same fan profiles hiding undermonetized elsewhere in the data? This gives judges a concrete reference before the analytical findings land.
Pre-work split:

- **Tucker:** Premise deep-dive (for/against cases), guiding questions to steer day-of analysis.
- **Khalid:** Outlier research + technical prep (EDA template, clustering pipeline, Streamlit dashboard, presentation skeleton).

Day-of principles:

- Demographics as default until data confirms psychographic story — then upgrade
- Presentation done by 3 PM, multiple rehearsal runs
- 2:45 PM hard stop on coding (non-negotiable pact with Tucker)
- If you advance, you can polish the deck for conference stage

The Analytical Framework

Your analysis follows a three-act structure:
Act 1 — Segmentation reveals hidden fan archetypes. Cluster fans into 3–5 segments. Give each a vivid identity. Prove "women's sports fans" is not a monolith.

Act 2 — The Commercial Value Index exposes the monetization gap. A composite metric per segment combining spending, brand receptivity, engagement depth, social amplification, and growth potential. Show that certain segments generate disproportionate value that "revenue per fan" misses. The iso-fan who spends heavily online? The Gen Z values advocate who amplifies brands socially? Invisible to legacy analytics.
Act 3 — New partnership models unlock the untapped value. For each high-value segment, propose a specific activation strategy. Be concrete: "Segment 3 indexes 2.3x on social engagement and 1.8x on brand recall — a streaming-native partnership with interactive second-screen experiences captures $X in incremental value."

The Commercial Value Index (CVI)

This is your analytical innovation — the thing that directly answers "beyond revenue per fan."
Formula:

CVI = 0.30(Spending) + 0.25(Brand Receptivity) + 0.20(Engagement Depth)
+ 0.15(Social Amplification) + 0.10(Growth Potential)

Each sub-score is normalized 0–1 using min-max within the dataset. On Thursday, you map actual columns to each sub-score. The structure is pre-built.
The "so what" chart: Plot segment size (% of population) vs. segment share of total CVI. Segments above the diagonal punch above their weight. Segments below are underperforming relative to size. This is the chart that makes judges lean forward.

Presentation Structure

Slide	Content	Time	Owner
1	Title + one-line thesis	15 sec	Tucker
2	The problem: "94% feel misunderstood" + monetization gap	45 sec	Tucker
3	Data overview + methodology (brief)	30 sec	Khalid
4	Finding 1: The fan segments (radar chart)	60 sec	Khalid
5	Finding 2: The undervalued segment / CVI reveal	60 sec	Khalid
6	Finding 3: Partnership model derived from segments	60 sec	Tucker
7	Recommendations: "What to do Monday morning"	45 sec	Tucker
8	Summary + Q&A	—	Both

The SCQA opening (Tucker delivers this):

- **Situation:** Women's sports fandom is growing explosively — $1.88B in revenue, 4.5x men's growth rate
- **Complication:** 66% of women fans feel sports organizations don't understand them; traditional revenue-per-fan models miss the real value

- **Question:** What does the data reveal about untapped commercial potential?
- **Answer:** [Your key finding — plugged in Thursday]

Persona naming convention: Evocative, memorable names beat "Cluster 1." Pre-loaded options to adapt on Thursday:

- "The Values Advocate" — highest on social values / cause alignment
- "The Sports Purist" — highest on game attendance / stats engagement
- "The Social Spectator" — highest on social media / group attendance
- "The Silent Superfan" — high spend, low social signals (iso-fan pattern)
- "The Casual Curious" — low frequency, high growth indicators
- "The Digital-First Devotee" — streaming/app heavy, low in-person
- "The Next-Gen Evangelist" — youngest, highest brand receptivity

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PART 6: THE DAY-OF BATTLE PLAN

Hour-by-Hour

Time	Activity	Khalid	Tucker
9:00–9:30	**THE REVEAL.** Read problem statement twice. Look at column names. Whiteboard hypotheses. DO NOT CODE YET.	Both	Both
9:30–10:30	**PARALLEL TRACKS.**	Run EDA template. Share findings every 15 min.	Draft narrative arc. Start slides 1–2.
10:30–10:40	**DECISION POINT.** "What's the ONE question we're answering?" Write it down.	Both	Both
10:40–12:30	**BUILD PHASE.**	Data cleaning → clustering → profiling → CVI. Sync at 11:30.	Populate slides 1–3. Design persona cards. Prepare recommendations framework.
12:30–1:00	**LUNCH + CHECKPOINT.** Do we have: clear answer, 60-second explanation, a "wow" moment? If missing "wow," pivot now.	Both	Both
1:00–2:30	**VISUALIZATION + INTEGRATION.**	Radar chart → CVI bar → scatter plot → (Streamlit if time). Export PNGs to slides.	Speaker notes. Narrative flow. Recommendations slide with $ figures. Q&A prep.
2:30–2:45	**FINAL ASSEMBLY.** All charts in slides. All notes written. Cut anything broken.	Both	Both

2:45–3:30	**REHEARSE.** Three runs. Timed. Cut anything over the limit. STOP ALL CODING.	Both	Both
3:30–3:45	**PRE-GAME.** Projector check. Close tabs. Water. Breathe.	Both	Both
3:45+	**PRESENT.** Lead with conclusion. Sound like you live here.	Findings + technical	Opening + recs + Q&A

Emergency Playbook

Problem	Solution
Clustering produces nonsense	Manual percentile segmentation on top 3 variance columns. Cross-tabs still tell a story.
Streamlit crashes	Static Plotly PNGs pasted into slides. Nobody will know.
Running out of time	Present what you have. Confident 70% beats panicked 100%.
Data is tiny or weird	Descriptive analytics + hypothesis generation. "Here's what this suggests, here's the study we'd design to validate." Judges respect honesty.
No clear clusters	Percentile-based segmentation (top 25%, middle 50%, bottom 25% on key metrics). Less elegant, still actionable.
Data has no psychographic columns	Demographics + behavior is your segmentation. Still valid. Mention you'd WANT psychographic data as a next step.

PART 7: THE STATS YOU MUST KNOW COLD

The Power Five (Drop These Like You've Known Them Forever)

#	Stat	Source	When to use it
1	**$2.5B** media rights opportunity by 2030	McKinsey	Opening: sizing the opportunity
2	**66%** of women fans say sports orgs don't understand them	Wasserman	Problem slide: the gap
3	**2x engagement, 21x less earnings** for women athletes	Wasserman/RBC	Athlete value arbitrage argument
4	**286% ROI** for WNBA Changemaker sponsors	Deloitte	Sponsor business case
5	**46%** of Gen Z more likely to buy from women's sports sponsors	Parity	Gen Z commercial case

The Supporting Cast (Use As Needed)

Stat	Source
$1.88B global revenue in 2024 (nearly doubled from $981M in 2023)	Deloitte
4.5x revenue growth rate vs. men's sports (2022–2024)	Deloitte
84% of women fans are key household purchase decision-makers	Wasserman
75% of discretionary spending controlled by women by 2030	Wasserman
18–21% of women's sports fans are iso-fans (vs. 5.6% in men's)	Wasserman
50% identify as fans of women's sports generally, not a specific team	Wasserman
WNBA media deal: $200M/year, 3x prior deal	Multiple
NWSL broadcast deal increased 40x	Multiple
WNBA attendance +48% YoY, NWSL +42%	Wasserman/RBC
WNBA sponsor value up 312% YoY	SponsorUnited
77% of Gen Z trust women athletes' endorsements	Parity
89% more inspired to take social action with women athlete sponsors	Wasserman
Angel City: $29 sponsorship revenue per social follower	Angel City data
Valkyries: <5% overlap with Warriors season ticket holders	GSW data
Women's NCAA final: more viewers, 4x cheaper ad spots than men's	McKinsey
Only 5% of SportsCenter airtime covers women's sports	USC study

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PART 8: THE TECHNICAL TOOLKIT

What's Pre-Built (In Your ssac-2026/ Folder)

File	Purpose
`notebooks/01_eda.ipynb`	Auto-EDA + manual exploration. Change filename, run all cells.
`notebooks/02_clustering.ipynb`	End-to-end clustering with manual override options.
`notebooks/03_analysis.ipynb`	Post-clustering deep analysis + CVI computation.

`scripts/preprocessing.py`	Data cleaning, type detection, encoding.
`scripts/clustering.py`	K-Means, K-Prototypes, K-Modes, HDBSCAN, auto-select.
`scripts/profiling.py`	Cluster profiling, naming, CVI computation.
`scripts/visualization.py`	Radar, scatter, heatmap, bar, bubble charts. Consistent palette.
`app/dashboard.py`	Streamlit interactive dashboard. Plug in data and go.

Clustering Decision Tree

```
Is data mixed (categorical + numeric)?
  YES → K-Prototypes (kmodes library)
  NO →
    Mostly numeric / Likert scales?
      YES → StandardScaler → K-Means → silhouette for optimal k
      High dimensionality (>15 features)?
        YES → PCA to 80% variance first, then K-Means
    Mostly categorical?
      YES → K-Modes or Latent Class Analysis (stepmix)
    Nothing works?
      → Percentile segmentation on top 3 variance columns
```

Visualization Priority (Thursday Build Order)

1. **Radar chart** — persona profiles (signature visual, goes on slides 4–5)
2. **CVI bar chart** — segments ranked by commercial value (goes on slide 5–6)
3. **Size vs. value bubble chart** — who punches above their weight (the "so what" chart)
4. **UMAP/PCA scatter** — cluster separation proof (methodology slide)
5. **Heatmap** — over/under-indexing by segment (supporting detail)
6. *If time:* Streamlit interactive dashboard (the "wow" demo)

PART 9: FLASHCARDS (QUIZ YOURSELF)

Cover the answers. Say them out loud. If you can nail 80%, you "know hockey."

Q: What is women's sports global revenue in 2024? A: $1.88B (nearly doubled from $981M in 2023).
Q: What's the biggest revenue stream and what share? A: Commercial/sponsorship at ~54%.
Q: McKinsey's media rights opportunity figure? A: $2.5B by 2030.
Q: What % of women fans feel sports orgs don't understand them? A: 66%.
Q: What's the iso-fan prevalence in women's vs. men's sports? A: 18–21% vs. 5.6%.
Q: What % identify as fans of women's sports generally (not team-specific)? A: 50%.
Q: WNBA broadcast growth projection? A: +32%.
Q: Valkyries expansion fee → current valuation? A: $50M → ~$500M (10x in under 2 years).
Q: Gen Z purchase lift for women's sports sponsors? A: 46%.
Q: Women athletes' engagement vs. earnings comparison? A: 2x engagement, 21x less in salary.
Q: WNBA Changemaker sponsor ROI? A: 286%.
Q: Angel City's revenue per social follower vs. Dodgers? A: $29 vs. $13.

Q: What % of women sports fans are key purchase decision-makers? A: 84%.
Q: Nebraska volleyball world record attendance? A: 92,003.
Q: Name the three Wasserman fan archetypes. A: IsoFan (18–21%), DuoFan (25–28%), Social Fan (54–59%).
Q: What's the CVI formula weighting? A: Spending 0.30, Brand Receptivity 0.25, Engagement 0.20, Social Amplification 0.15, Growth Potential 0.10.
Q: Four adjacent market buckets from New Economy Part III? A: Capital vehicles, property/high-growth sports, performance tech, media/fan ecosystems.

PART 10: YOUR 60-SECOND "WE KNOW WOMEN'S SPORTS" OPENING

Memorize this. Adapt it Thursday with real numbers.

> "Women's sports fandom is the most commercially misunderstood audience in sports. Revenue nearly doubled to $1.88 billion in 2024 — growing 4.5x faster than men's — but the monetization gap is actually widening: women's media revenue per viewing hour is declining while men's grows. At the fan level, 84% of women fans drive household sports purchases, yet 66% say sports organizations don't understand them. Our analysis reveals [X] distinct fan segments, each with different motivations, behaviors, and commercial potential. The segment we call [NAME] represents [X]% of the fan base but accounts for [X]% of our Commercial Value Index — and they're almost completely untouched by current partnership models. Here's how to reach them."

If you can say that without looking at notes, you've already beaten 80% of the room.

EPILOGUE

You have 21 hours of prep and 7 hours of execution. You have a teammate who's been in this room before. You have a technical toolkit that automates the first 90 minutes of analysis. You have deeper domain knowledge than most people who work in women's sports business development. And you have a narrative framework that speaks Wasserman's exact language.

The teams that lose will either: (a) still be coding when they should be rehearsing, (b) present methodology instead of insights, or (c) fail to answer "so what would you actually do Monday morning?"

You're not going to do any of those things.

Because you read the log. You built the tools. You know the stats. And you're going to science the shit out of this.

Sol 1. All systems nominal. Let's go.